



Work Experience

Centre for Development of Telematics (DoT, GoI) – Low-Latency Systems Engineer (Scientist 'B')

Jun'24 - Present

National Scale Low-Latency & High-Throughput Packet Processing Solution

- Designed and engineered a **DPDK-VPP kernel-bypass packet processing engine** in C sustaining **400 Gbps line-rate Rx/Tx** on a single commodity server, leveraging VPP's vector-batch **cache-warm execution** model and DPDK PMD busy-poll I/O loops to achieve deterministic **sub-microsecond** forwarding latency — directly analogous to co-location-grade market-data feed ingestion and tick-to-trade fast paths.
- Engineered **lock-free, wait-free** ring buffers (rte_ring) and hugepage-backed memory pools (rte_mempool) to eliminate dynamic heap allocation in the fast path, achieving **zero malloc contention** under sustained 400 Gbps load — identical to the zero-alloc constraint in HFT order management hot paths.
- Implemented **CPU affinity pinning**, **NUMA-local hugepage allocation**, and **cache-line-aligned struct** layouts, isolating forwarding threads to dedicated physical cores and eliminating both cross-socket memory latency and false sharing — reducing L1 D-cache miss rate by **~40%** and delivering **jitter-free, deterministic** throughput.
- Engineered **NIC hardware offloads** (RSS, checksum offload, flow classification) combined with **AVX-512 SIMD intrinsics** for bulk packet classification, achieving a **10× throughput uplift** and **15× CPU utilization reduction** — directly translating to NIC-level market-data timestamping, hardware filtering, and SIMD-accelerated protocol decoding in co-location environments.

Stateful DPI Pipeline & Protocol Analysis at Line Rate

- Implemented C-based Deep Packet Inspection (DPI) for **eSNI extraction** from UDP/TLS (**QUIC**) flows by parsing TLS ClientHello extensions via **nDPI** — demonstrating **byte-level protocol parsing** at wire speed, directly applicable to FIX/ITCH/OUCH market-data protocol parsing in HFT feed handlers.
- Engineered a **two-phase Producer-Consumer stateful pipeline** across RSS-affined threads, with Producer and Consumer co-pinned to the same physical core for **L1/L2 cache sharing** — enabling near-zero-latency **packet handoff** while maintaining per-flow state across correlated packets, mirroring HFT order tracking and position management hot paths.

Autonomous AI-Powered Low-Latency Development Infrastructure

- Architected and deployed a **fully air-gapped LLM inference stack** on Dell PowerEdge R760 (NVIDIA A100 80GB PCIe) using vLLM 0.21 + Qwen3.6-27B-FP8 with RadixAttention KV-cache prefix caching (43GB pool, ~675K token capacity), achieving **~60 tok/s throughput** and **sub-50ms repeated-query latency** for real-time IDE assistance with **zero data egress** — directly applicable to proprietary strategy and code confidentiality requirements in HFT.
- Built a hybrid **RAG pipeline** using Qdrant vector DB with BAAL/bge-m3 (1024-dim) **dense + sparse** embeddings, ingesting DPDK/VPP API references, Intel AVX-512 manuals, internal documentation, and community resources via automated nightly cron jobs — ensuring always-current retrieval-augmented context for dev team.
- Developed a Python **MCP server** (AsyncOpenAI, FastMCP, Qdrant) with @file context selection, web search, URL ingestion, and Mermaid.js doc generation integrated into VS Code/Cursor via **STDIO**; designed a **LangGraph-based autonomous CI/CD pipeline** performing Clang/Tree-Sitter AST analysis, gcc self-correction loops, and DPDK standards compliance scanning (rte_malloc enforcement, cache-alignment verification) — cutting code review turnaround from hours to minutes.

York University (Toronto, Canada) – Cybersecurity and AI-ML Researcher (MITACS Research Intern)

Jun'23 - Aug'23

Automated Intelligence-Driven Malware Detection and Analysis

- Implemented **Artificial Intelligence & Machine Learning techniques** like **information-gain selection** and **three-layer Random Forest** architecture plus **weighted score fusion core** for multi-source behavioural analysis over network flows and memory dump features.
- Developed** an automated **Python + QEMU Hypervisor testbed**, orchestrating thousands of concurrent instances of Windows 10 VM which has been “aged” for one week to defeat sandbox evasion; and then 2000 malware samples are executed across 8 families, generating a **17TB database** of pcaps and memory dumps.
- Performance:** L1 (network binary) macro **P/R/F1 ≈ 0.96/0.96/0.95**; L2 (network family) **macro F1 ≈ 0.98** with ~15% Unknown; L3 (memory family) **macro F1 ≈ 0.98**.

VolMemLyzer-V2

- Designed and developed** a feature extractor tool – **VolMemLyzer - V2** with **250+** supported **Memory Forensics Analytics Features** using **Volatility3** framework.
- Implemented** modular pipeline which reduces processing duration of stored memory snapshots by **20–30%**, and producing CSVs for downstream **Machine Learning**.

Technical Skills, Corporate Trainings & MOOCs

Programming/Scripting		C/C++, Python, Bash, PowerShell, JAVA, MATLAB	Trainings/ MOOCs	♦ Hacking Android, iOS, and IoT apps by example by c0c0n ♦ High Perf. Computing & Linear Prog. by IISc Bangalore* ♦ Introduction to Digital Forensics by Security Blue Team ♦ Data & Tools for Defense Analysts by Splunk STEP
Low-Latency / HFT Systems		Kernel Bypass (DPDK/PMD), Lock-Free & Wait-Free Data Structures, CPU Pinning & Core Isolation, NUMA-Aware Allocation, Hugepages, TSC-Based Timestamping, Busy-Poll I/O, Cache-Line Alignment, AVX-512/AVX-2 SIMD, NIC Hardware Offloads, Zero-Copy Networking		
AI/LLM Infrastructure		vLLM (RadixAttention, Prefix Caching, Speculative Decoding), LangGraph (Autonomous Agents), Qdrant Vector DB, RAG, MCP, LLM Quantization, Local LLM Deployment	Networks	DPDK, VPP (Vector Packet Processor), nDPI, TCP/IP Stack, BGP, DNS RPZ, QUIC, TLS/eSNI Parsing, FIX / ITCH / OUCH (market data protocols — familiarity), ICMP, OSI Model
AI-ML Techniques		CNN, RNN, Random Forest, XGBoost, SVM, KNN, PCA		
Cybersecurity		Volatility, Metasploit, Wireshark, NMap, YARA, BurpSuite, Autopsy, Splunk, Suricata, Frida, Magisk, Xposed, Jadx, R2	CI/CD Tools	Docker, Docker Swarm, Github Workflows, QEMU, VMware Tools integration, Kubernetes, Playwright

* Under Training

Education & Certification

Degree	Institute	Grade	Remarks	Year
B. Tech. (CSE)	Indian Institute of Technology, Jodhpur	7.88/10.0	Institute Silver Medal	2020-2024
AISSCE (Class XII)	Army Public School, Birpur	93.80%	Top 0.1% nationally	2018-2019
Certifications			Issuing Body	Year
Certified Ethical Hacker v13 (Training Completion Certificate)			EC Council	2026*
CompTIA Security+ (SY0-701) + Google Cybersecurity			CompTIA	2025
Certified in Cybersecurity			ISC2	2024

* Employer Sponsored (Exam Pending)

Publications

[Unveiling Evasive Malware Behavior: Towards Generating a Multi-Sources Benchmark Dataset and Evasive Malware Behavior Profiling Using Network Traffic and Memory Analysis](#) | [The Journal of Supercomputing](#) (Springer)

Achievements

- Received **Student Distinguished Services Award** and **Institute Silver Medal** for my contributions to Student Senate IITJ in AY 2022-23.
- Contingent Leader** of 150 IIT-J students in the 55th INTER IIT SPORTS MEET 2022; personally led Runner-Up awardee **Best Marching Contingent** at IIT Delhi.
- Achieved **All India Rank 1** in Microsoft Cybersecurity Engage 2022.